MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First semester

Multi-Disciplinary Course-MG1MDCFSQ100-FOOD PROCESSING TECHNOLOGY

(2024 ADMISSION ONWARDS)

QP CODE: 24900106 Max marks:35

Part A

Multiple Choice Questions

(Answer all questions/each question carries 1 mark)

1. Evaluate the significance of high-pressure processing (HPP). [CO 3=1]
   Ans: b) it preserves food without heat
2. In the 20th century, which industry's growth is closely linked to large scale food processing. [CO 1=1]
   Ans: a) Fast-food industry
3. Recognise the process that helps in eliminating harmful microorganisms from food. [CO 3=0.5, 4=0.5]
   Ans: a) Sterilization
4. Examine why canned foods are popular in remote areas. [CO 1=0.5, 3=0.5]
   Ans: b) they last longer without refrigeration
5. Compare the main difference between primary and tertiary processing. [CO 1=0.6, 3=0.2, 4=0.2]
   Ans: b) primary is for raw materials, tertiary for consumer-ready products
6. Choose a method which involves cooking food directly over an open flame? [CO 1=0.5, 3=0.5]
   Ans: c) grilling
7. Select the process for improving the tenderness of tough cuts of meat. [CO 1=0.5, 3=0.5]
   Ans: c) Braising
8. Identify the type of drying where moisture in food is frozen and sublimed to vapour under vacuum. [CO 1=0.5, 3=0.5]
   Ans: c) Lyophilization
9. Select the drying method that uses high temperatures to evaporate moisture quickly. [CO 1=0.4, 3=0.4, 4=0.2]
   Ans :b) hot air dying
10. Recognise a potential concern regarding food irradiation. [CO 1=0.5, 3=0.5]
    Ans : c) May cause the formation of harmful compounds
11. Select the name which suits to food irradiation. [CO 1=0.5, 3=0.5]
    Ans: a) Radurization

12. Name the method that involves storing food just above its freezing point.

[CO 1=0.5, 3=0.5]

Ans: b) refrigeration

13. Identify the primary goal of freezing food. [CO 1=0.2, 3=0.6, 4=0.2]

Ans: b) retarding microbial growth

14. Select the maximum shelf life of most foods in a freezer. [CO 1=0.5, 3=0.5]

Ans: c) 1 Year or more depending on type of food

15. Name the term used for formation of ice crystals in frozen food.

[CO 1=0.5, 3=0.5]

Ans: b) Crystallisation

Part B

Short essay type questions

Answer ant 4 questions/each question carries 5 marks

16. Discuss how freezing preserves the nutritional value of food.

Ans: [CO 1=1, 3=2, 4=2]

- Freezing definition, temperature range
- benefits of freezing
- freezing methods

17. Discuss the importance of storing perishable food at low temperatures.

[CO 1=1, 3=2, 4=2]

Ans:

- perishable foods- define
- low temperature preservation methods
- importance of low temperature methods

18. Define drying. Describe the various methods of drying. [CO 1=1, 2=2, 3=2]

Ans:

- Drying – definition, purpose
- Methods of drying (sun drying, mechanical drying), types of dryers

19. Discuss the advantages and disadvantages of drying as a food preservation method. [CO 2=2.5, 3=2.5]

Ans:

Advantages	Disadvantages
Prevents spoilage	Potential contamination
Extends shelf life	Slow process
Cost-effective	Exposure to oxygen
Reduces packaging, transportation, and storage costs	Needs attention

	Use of Mechanical dryers require skilled labours to operate
	Maintenance is required for dryers

- Advantages and disadvantages of different types of dryers can be included.

20. Compare traditional and modern methods of food preservation. [CO 1=2.5, 3=2.5]
    Ans:
    - Food preservation – definition
    - Importance/need of food preservation
    - Different methods of preservation – traditional and modern methods– advantages, disadvantages, applications

21. Discuss the advantages and disadvantages of steaming as a cooking method.

[CO 1=1, 2=2, 3=2]

Ans: Steaming – definition, applications

***Advantages*:**

- It preserves the main nutrients **-** As there is no direct contact with water, the nutritional substances in food are not lost. Food therefore retains its nutrients much better.
- Full and genuine flavour **-** Steamed foods retain their own genuine flavour, allowing us, in a certain sense, to "rediscover" their real essence. This cooking method is ideal for fish and vegetables in particular.
- Less chance of burning and scorching.
- Texture of food is better as it becomes light and fluffy.
- Almost no smells **-** In most cases, steam cooking reduces the unpleasant smells typical of some foods.
- Cooking time is less and fuel wastage is less.
- Steamed foods like idli and idiappam contain less fat and are easily digested and are good for children, aged and for therapeutic diets.
- Nutrient loss is minimised.
- Quick and inexpensive **.**

Disadvantages:

- Steamed vegetables can taste bland: addition of seasonings can overcome this blandness.
- Constant attention is needed to avoid overcooking.
- Steamers use a lot of water.
- Steaming equipment is required.
- This method is limited to the preparation of selected foods..